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One Acre, Zero Dependency!

The starting points of this project are, quite literally, two dots on two different maps. Now both are just open fields, a blank canvas so to say.

Over the coming year (2026/2027) they become the testing grounds for something more ambitious: an off-grid, AI-driven life built the way it probably should have been built. In rhythm with nature, not in spite of it.

I'll be experimenting with solar, composting humanure, waste management, rainwater harvesting, and a garden grown on Rudolf Steiner's biodynamic sowing calendar. But the underlying model I'm borrowing isn't European it's older, and it comes from the Great Plains. Long before "off-grid" was a lifestyle category, Native American tribes moved with the seasons: an open summer camp for grazing, hunting, and growing, and a sheltered winter camp for protection and warmth. They didn't fight the seasons; they migrated with them and treated that migration as the infrastructure that made a sustainable life possible.

I'm taking the same approach, minus the horses. I live a mobile, low-impact life out of a self-sufficient van leaving no permanent trace, farming one site through summer and the other through winter, moving as the season and the biodynamic calendar demands.

AI is what makes that kind of mobility manageable for one person instead of a whole tribe: models tracking the calendar, the soil, the sky, the batteries and the harvest.

This raises a question well worth asking, and one that will be underlining the series:

In modern Europe, do we point AI at agriculture to streamline and harvest ***faster*** Squeezing more certainty and yield out of the land, regardless of what the season is doing? Or do we use AI to read the land and sky more closely and move ***with*** nature the way seasonal migration always did?

I've spent many years in enterprise IT and cloud sales, selling infrastructure to companies that never once had to think about where their water or power actually came from. This project sits at the intersection of both halves of my life: hands in the soil, and a laptop full of models tracking what the soil, the sky, and the batteries are doing.

This article is the prelude to a series. Over the coming year I'll document, month by month and in step with the biodynamic calendar, how the land, the systems, and the AI tooling around them actually get built, the models, the training data, the failures, all of it as an open portfolio. This first piece lays out *why* I'm doing it this way, and *why now*.

Why now: a summer that made the case for me

I didn't need to invent a justification for reducing my dependence on centralized systems. 2026 has been supplying the evidence all year. Everywhere you look, the things happening around us that impact us directly are staggering.

Drought and water stress across Europe. This summer has been one of the most significant heat-and-drought events Europe has recorded in years. Between April and June, rainfall was below average almost everywhere, from Portugal to southern Finland, and the resulting drought has hit rivers, farms, and drinking-water systems hard.

Wildfires linked to the dryness have already burned well over 100,000 hectares in Spain and more than 40,000 in France. In the Netherlands, national water managers reported the lowest levels ever measured on the Rhine, disrupting shipping and letting seawater intrude into freshwater used for irrigation.

In Italy's Po basin, dozens of towns have needed drinking water delivered by tanker truck. Central Europe, Czechia, Slovakia, Hungary, exactly the region my own plot sits near is described by climatologists as facing droughts that will keep getting more frequent and more severe. Across the EU, electricity cooling alone eats up a third of all water use, agriculture another third a system with almost no slack left in it.

Deforestation and a warming baseline.

The UN's newest global forest assessment shows deforestation has slowed over the past decade but is still running at roughly 11 million hectares a year. With forest degradation and fire increasingly picking up where clear-cutting leaves off.

2024 was the worst year on record for tree-cover loss overall! Driven heavily by wildfire nearly 14 million hectares burned that year alone, with Russia and Canada worst affected. The world is still tracking well short of the reduction needed to end deforestation by 2030. Every one of those hectares was doing quiet work: holding water in soil, cooling regional climate, buffering the exact kind of heat-drought feedback loop now baking Europe.

A war that briefly closed one of the world's most important shipping lanes.

Since February 2026, the conflict between the US/Israel and Iran has run through strikes, a leadership assassination, a ceasefire, and repeated flare-ups around the Strait of Hormuz, a route that a huge share of the world's oil and gas physically has to pass through.

Iran has repeatedly threatened and attacked commercial vessels there, and by early August the two sides were still negotiating the terms of reopening it. None of that is abstract for anyone paying an energy bill. A regional war, six time zones away moved fuel prices and shipping insurance costs for everyone, instantly, because the systems we all live inside are so tightly coupled.

Put those three things next to each other and a pattern shows up that has nothing to do with any one crisis: **modern life is a long chain of dependencies on rainfall patterns that no longer behave, on forests that keep disappearing, on shipping lanes that one regional conflict can choke and almost none of us have any control over that chain ourselves.** Most households, mine included, have no independent water source, no stored energy, no food buffer, and no way to keep functioning if any single link breaks. We've optimized for efficiency and centralization for decades, and efficiency is exactly what turns a local shock into a global one.

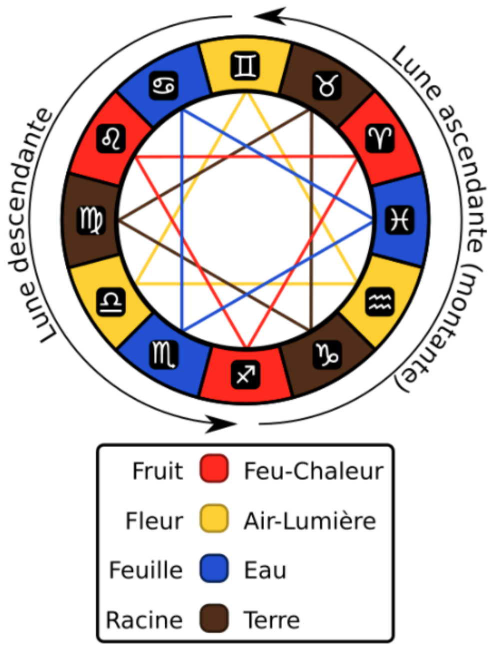
This is the case for my acre, not survivalism just moving a meaningful share of my own water, energy, and food supply onto a chain I control. Building it well enough that it’s not a downgrade in comfort or reliability.

AI is part of the solution, humans do the rest

Homesteading and off-grid living are old disciplines the Amish, generations of subsistence farmers, and the biodynamic movement itself have been doing versions of this for a century or more without a single sensor. What's changed is that a single person, working alone on one acre, can now use tools that used to require an entire research institute. That's the actual argument for AI here: not novelty, **but leverage**. A few concrete examples that are fit for AI:

1. Tracking the biodynamic calendar and building the planting schedule

Steiner's biodynamic calendar organizes farming activity around the Moon's position against the zodiac and the rhythms of the planets root days, leaf days, flower days, fruit days layered on top of ordinary seasonal planting logic. Doing this by hand means constantly cross-referencing an almanac against your own microclimate, soil temperature, and frost dates. That's a small, well-defined data problem: a model that ingests the calendar's root/leaf/flower/fruit cycle, my specific latitude and soil data, historical frost and rainfall records for the region, and each crop's own timing needs, and turns it into an actual week-by-week planting and harvesting schedule with reminders rather than a chart I have to reinterpret every few days.



source: Wikipedia

2. Modeling power: production, storage, and consumption together

Off-grid solar lives or dies on the gap between what the panels make and what the household uses, hour by hour, and lead-acid or LiFePO4 batteries only forgive so much mismatch before they degrade. This is now a well-studied machine learning problem: recent research shows ML-based solar forecasting models (LSTM and similar time-series approaches) cutting forecast error by 15–20% over older statistical methods, using weather and irradiance data alone. Pairing a forecast like that with a simple reinforcement-learning dispatch policy deciding when to run the water pump, the workshop tools, or the well pump based on predicted sun rather than current sun is the difference between oversizing a battery bank "just in case" and running a right-sized system with confidence. Separately, researchers have shown that battery health itself can be predicted from ordinary field data (voltage, current, temperature logs) with over 70% accuracy on end-of-life. AI can predict this more accurate, which matters enormously when the nearest replacement battery isn't a same-day Amazon order.

3. Biofertilizer management

Composting humanure, garden waste, and liquid manure (gier/Jauche) safely and effectively is mostly a fermentation and nutrient-timing problem: temperature, carbon-to-nitrogen ratio, moisture, and time all have to line up, and getting it wrong either wastes the fertilizer or creates a health hazard. Cheap sensors (temperature, moisture, pH) feeding a small model can flag when a compost pile needs turning, estimate when a batch is safe and ready, and cross-referenced against the biodynamic calendar and each bed's actual nutrient history. Recommend what to apply, where, and how much, instead of me guessing from a logbook.

Credits: istockphoto

4. Food inventory and storage

A pantry, root cellar, and preserved-food store are small warehouse problems: what's in stock, what's about to spoil, what needs to be eaten, canned, or restocked. (before the next harvest window). A basic vision model (a photo of a shelf) or even a manually logged inventory feeding a simple tracking system can flag expiry windows. The Model can suggest what to cook or preserve next. Forecast using this year's actual harvest data if the garden's yield will cover the household through winter. Incorporating enough lead time to fix a shortfall before it becomes a crisis rather than after.

None of these are exotic uses of AI. They're the same categories time-series forecasting, anomaly detection, simple recommendation systems used across industrial agriculture and grid operations, just re-pointed at a two-acre plot and one household instead of a utility company or an agribusiness. That re-pointing is the actual project, making these tools accessible for anyone to use and implement.

What "the way forward" means here

I'm not arguing everyone should buy land and go off-grid that’s neither realistic nor necessary for most people. The argument is narrower: the systems most of us depend on for water, energy, and food are more centralized and more exposed to distant shocks than most of us think about.

This summer's drought, the Iran war, Ukraine war, forest fires around Europe, extreme temperatures are just very legible reminders of that. Building even partial independence from those systems and using AI to make that independence *manageable* for one person or a family.

This is not AI fiction anymore, the technology is available, data is available and can be provided in a mobile or desktop/online application. The possibilities are endless when thinking about the different implementation options.

That's what the rest of this series will show, in public, as I build it: the biodynamic planting model, the energy dashboard, the compost and gier tracker, the inventory system code, data, and mistakes 😉 included.

Next in this series: *mapping the biodynamic calendar to my actual planting plan, and the first version of the planting-schedule model.*

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